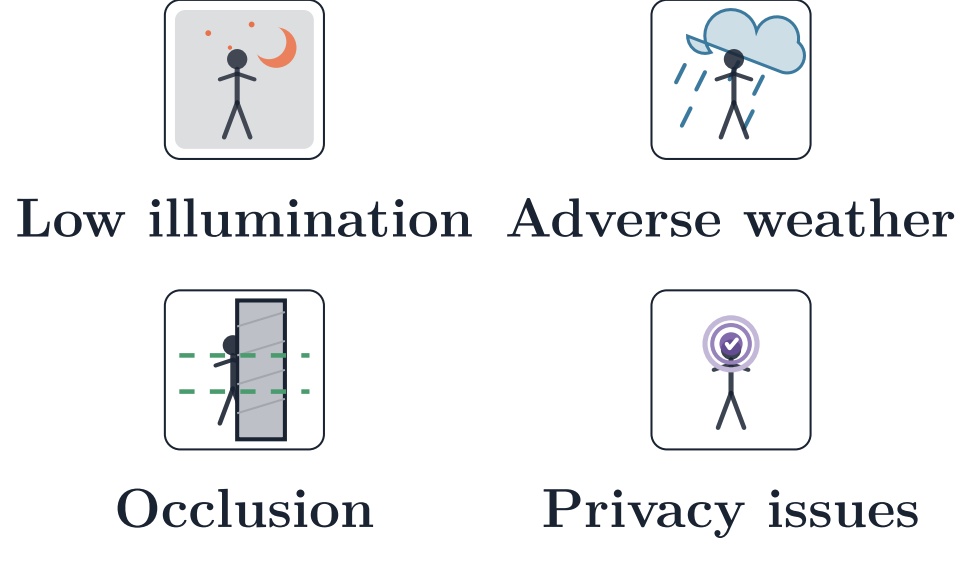


Few-Shot Radar Person Identification via Tracking-Integrated InISAR and Pre-Trained Meta-Learning

Jeffrey Hsiung* Rengin Torun* S. Hamed Javadi† Hichem Sahli†
KU Leuven | imec | Vrije Universiteit Brussel *Equal contribution (first authors) †Equal contribution (supervisors)

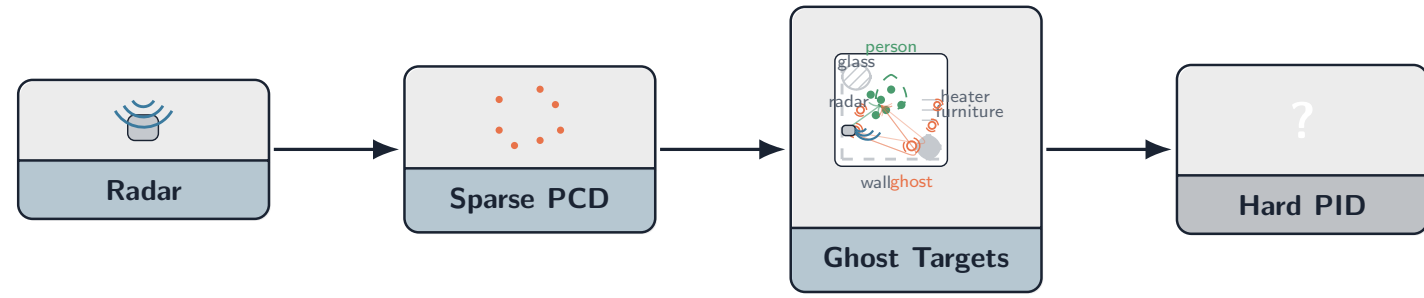
1 Introduction

Vision-based person identification (PID) struggle under conditions [1, 2]:

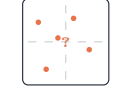


Radar Advantages: Zero optical exposure · All-weather operation · Penetrates thin barriers · Privacy preserving

2 Challenges in Radar-Based PID



Radar point clouds are sparse, noisy, and contaminated by multipath ghosts



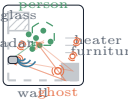
Low Angular Resolution

Conventional angle-of-arrival [3] yields motion-defocused clouds and multipath ghost artifacts.



Rigid-Target Bias

Interferometric Inverse Synthetic Aperture Radar (InISAR) improves resolution but is validated only on large *rigid* targets [4, 5].



Multipath Ghosts

Ghosts overlap in range-Doppler and mimic real 3D clusters [6]; indistinguishable for DBSCAN.



Data Scarcity

Existing SOTA [7, 8] demand > 100 sequences/person; Point Cloud Transformer (PCT) [9] handles spatial structure but targets *static* objects [10].

Gap: Tracking-integrated InISAR + pre-trained meta-learning.

3 Key Results

73.8% InISAR IoU
< 5.5 cm Height Error
70.0% Cross-Session PID

Contributions & System Impact



Articulated Validation: First InISAR on humans in multipath-rich indoor environments.



Resolution-level Error: < 5.5 cm (median 3.87 cm) approaches $\Delta r=4.55$ cm limit (vs. prior 0.2–1.6 m errors).



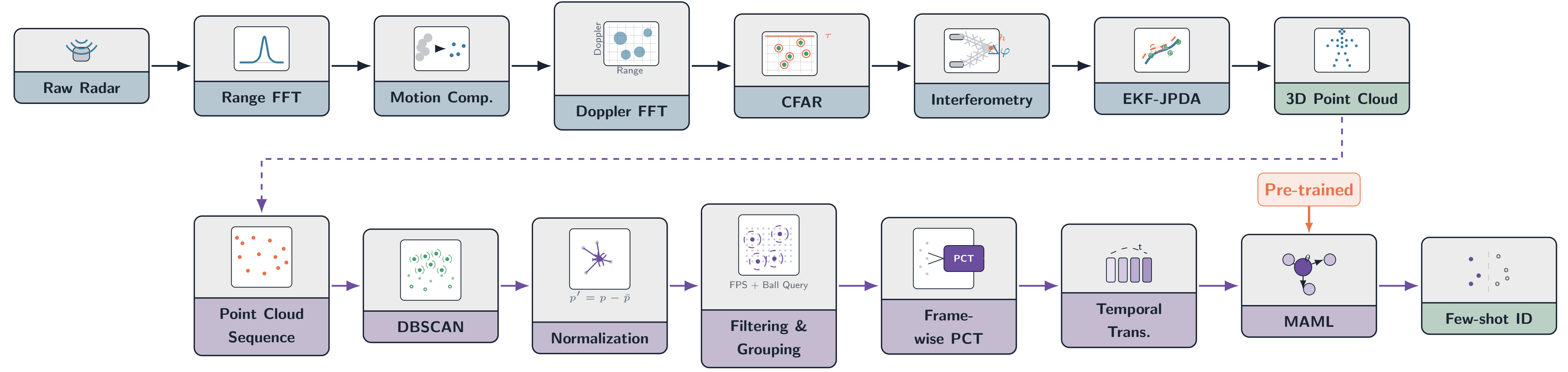
SOTA Supervised: PCT-T achieves 98.99% supervised (+2.81pp over TCPN).



MAML Gains: Pre-trained MAML gives 70.0% cross-session (+15.6pp over random).

Deployable few-shot radar PID with minimal training data.

4 Our Framework: Unified Sensing + Meta-Learning



End-to-end system: InISAR 3D reconstruction $\rightarrow$ spatio-temporal learning $\rightarrow$ PID

(1) Radar domain

Tracking-integrated InISAR reconstructs high-res 3D point clouds of articulated human motion, validated against motion-capture ground truth.

(2) ML domain

Spatio-temporal transformer (PCTtrans) embedded in MAML; task-specific pre-trained initialization substantially improves few-shot adaptation.

(3) System integration

High-quality 3D reconstruction + meta-learning together enable practical few-shot radar PID, reducing dependence on large subject-specific datasets.

5 InISAR Pipeline: 3D Reconstruction

Step 1: Range FFT

- FMCW chirp compression to range profile $S_{R,m}(f, t)$. Physical range limit $\Delta r = c/(2B)$ bounds our achieved height error.

Step 2: Motion Compensation (ICBA)

- Translational motion model [5, 11]:

$$R_0(t) = r_0 + \beta t + \gamma t^2$$

- Nelder-Mead optimizes image contrast $[\hat{\beta}, \hat{\gamma}]$ to estimate global translation, correcting smearing caused by human motion.

$$[\hat{\beta}, \hat{\gamma}] = \arg \max \left\{ \frac{1}{N} \sum (I_i - \bar{I})^2 \right\}$$

- Yields signal-to-noise ratio (SNR) enhanced range profile $S_{C,m}(f, t)$.

Step 3: Coherent Combination & Doppler FFT (ISAR Image)

- Geometric phase corrections align all MIMO virtual antennas (SNR boost) $\rightarrow$ focused 2D ISAR Range-Doppler Profiles (RDP).

Step 4: CFAR Segmentation

- SOCA-CFAR ($N_T = 6$, $P_{FA} = 10^{-10}$) isolates the target cells from noise/clutter.

Step 5: Interferometry

- Phase difference between coherently combined top/bottom MIMO rows yields elevation:

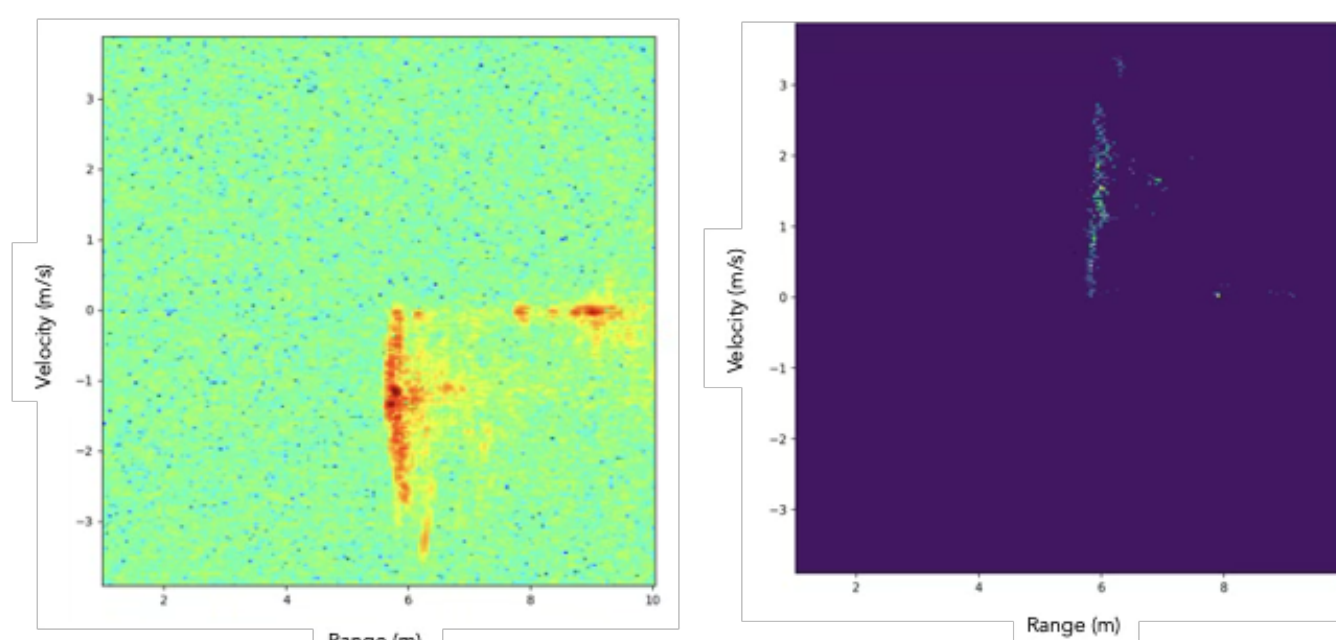
$$I_1 \cdot I_2^* = |I_1| |I_2| \exp \left(j \frac{2\pi}{\lambda} \Delta R \right)$$

$$\phi = \frac{2\pi}{\lambda} L \frac{h}{R}, \quad h = \frac{\lambda \phi R}{2\pi L}$$

- Exploiting MIMO antenna pairs reduces measurement noise.

Step 6: EKF-JPDA Tracking

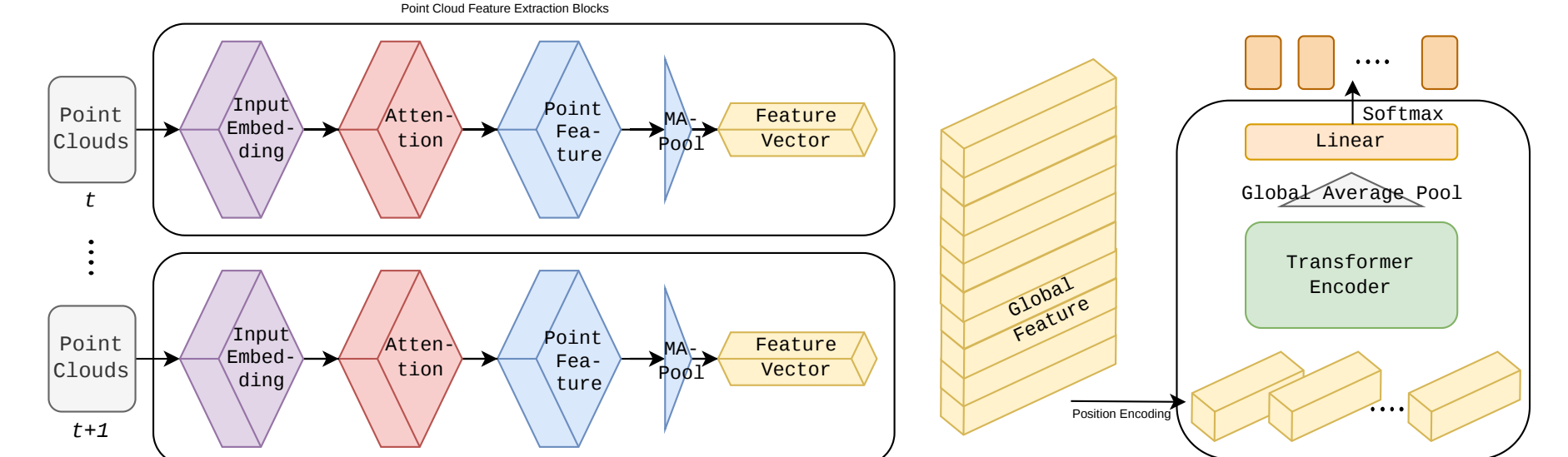
- JPDA handles data association; EKF filters state covariance.
- Max miss count=1 prunes multipath ghosts while tolerating occlusions.



Raw RDP

Post-InISAR-pipeline RDP

6 Few-Shot PID with PCTtrans



PCTtrans (PCT-T): frame-wise PCT + Temporal Transformer

Postprocessing:

- DBSCAN ($k = 8$, $S = 1.0$) + near-field filter
- Centroid normalization: $p'_i = p_i - (\bar{x}, \bar{y}, 0)$
- Random padding to $n_{\max} = 1000$
- Output: $\mathcal{S} = \{\mathcal{P}_t\}_{t=1}^{10}$, $\mathcal{P}_t \in \mathbb{R}^{1000 \times 5}$

Spatial (PCT Encoder [9]):

- Sampling-and-Grouping (SG): SG1: Farthest Point Sampling (FPS) $n_1 = 200$, $r_1 = 0.2$; SG2: FPS $n_2 = 100$, $r_2 = 0.3$
- Neighbor Embedding: $\Delta F(p) = \text{concat}_{q \in \text{kmn}(p,k)} (F(q) - F(p))$
- Offset-Attention: $F_{\text{out}} = \text{LBR}(F_{\text{in}} - F_{\text{sa}}) + F_{\text{in}}$
- Output: $\mathbf{f}_t \in \mathbb{R}^{192}$

Temporal (Transformer):

- Positional encoding: $\mathbf{f}_t^{\text{pos}} = \mathbf{f}_t + \text{PE}_t$
- Multi-head self-attention ($h = 4$)
- Max + average pooling $\rightarrow \mathbf{g} \in \mathbb{R}^{192}$
- Classification:

$$p(y|\mathcal{S}) = \text{Softmax}(W_2 \cdot \text{ReLU}(W_1 \mathbf{g} + b_1) + b_2)$$

Key Design Choices:

- FPS + Ball Query handles non-uniform density.
- Offset-Attention emphasizes local structure.
- Temporal transformer models gait dynamics.

8 Pre-trained Meta-Learning [12]

Two-Phase Algorithm:

Phase 1: Supervised training on 3 subjects $\rightarrow \theta_0^{PT}$

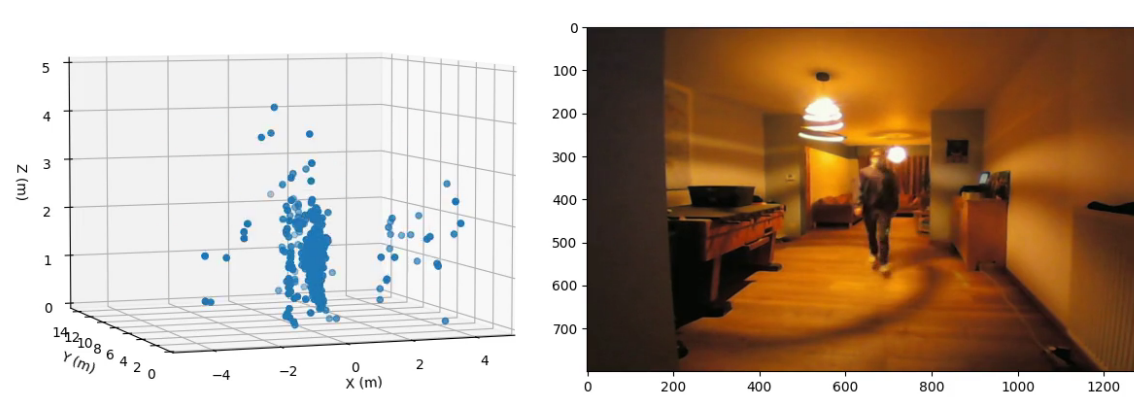
Phase 2: Model-Agnostic Meta-Learning (MAML) [13, 14] meta-training with θ_0^{PT}

Inner loop: $\theta'_i = \theta - \alpha \nabla \mathcal{L}$ ($\alpha=0.1$). Outer loop: $\theta \leftarrow \theta - \beta \nabla \varepsilon_i \mathcal{L}$ ($\beta=0.001$).

Gain: +15.6pp over random MAML

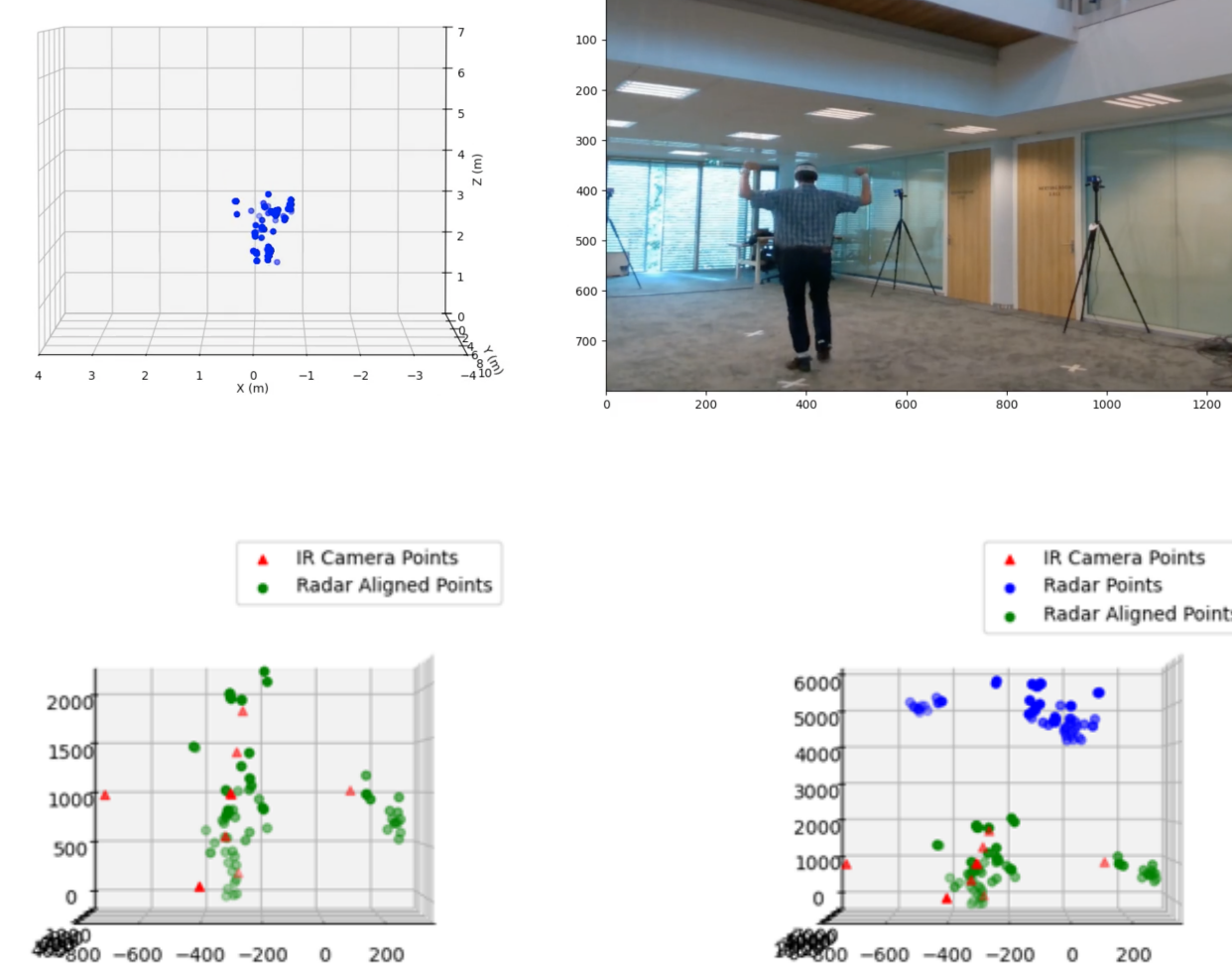
7 Experimental Results

Ghost Removal (EKF-JPDA)



(a) Before: Ghost remains

InISAR vs Motion Capture



Reconstruction Height Error < 5.5 cm (median 3.87 cm)

PID Benchmark Accuracy (%)

Query Type	TCPCN [8]	PCT-T MAML (sup.)	MAML (rand.)	MAML (pre-train)
Q1(same session)	96.18	98.99	76.67	94.44
Q2(cross-session)	35.10	32.05	54.44	70.00
Q3(cross-subject)	57.81	40.09	46.67	48.89

- Supervised:** PCT-T outperforms TCPCN $\rightarrow$ +2.81pp.
- Few-Shot:** Pre-trained MAML vs Random $\rightarrow$ +15.56pp.
- mmGait Benchmark:** PCT-T leads by +11.5pp on fixed trajectories [15].